PCA9615DP
DI2C IN
DI2C OUT
Conn_01x06
STM32L433CCU6
5077CR-16SMC2-BK-TR
USBLC6-2P6
AS5600 I2C (3V3/GND/SCL/SDA)
TB67S249FTG,EL
MOTOR A+/A-/B+/B-
PGND-GND STAR
Thermistor_NTC
SW_MEC_5G
LED

TERM: POPULATE AT BUS ENDS ONLY

AS5600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull-ups on the bus.

USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9: GPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_CR2.USV in firmware.
Internal D+ pull-up / termination. 90-ohm differential PCB pair.

STM32L433 IC4 <-> TB67S249 IC2 - stm32l433_tb67s249_connection.md

LO1/LO2/MO pull-ups: +3V3_D only (external digital 3.3V supply).
VM and +3V3_D supplied by power circuitry outside this specification.

AGC straps: R50=0; R52/R54/R56=100k; R51/R53/R55/R57=DNP.
CLIM1/BOOST/FLIM pull-ups must use DRV_VCC, never +3V3_D.

OSCM: R23=5.1k, C20=270pF; ~67kHz.
Place close to pin35. DRV_VCC: internal 5V, setting circuits only.

VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM-PGND loops short; paired pins joined at IC on PCB.

VREF: R29=10k series, R30=9.1k to GND, C21=100nF.
3.3V DAC full scale -> VREF 1.57V -> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.

NT1: single PGND-GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND, dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.

Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V: ENABLE Low or disable DAC for zero current.
Startup: check VM -> set DAC -> set DMODE -> wait >=7.5us -> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.

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